# Thirukumaran Kathirvelan

+91 7019214322 | thirukumaran3057@gmail.com | linkedin.com/in/kthiru44 | github.com/kthiru-44 | Bengaluru, India

*Research Interests: Quantum Information & Algorithms, Variational Quantum Circuits, Quantum Machine Learning*

## Education

**PES University** Bengaluru, India

*Bachelor of Technology in Computer Science and Engineering;* ***CGPA: 9.71 / 10.0***

- **Honors & Awards:** Prof. M.R. Doreswamy (MRD) Merit Scholarship (50% tuition distinction awarded to top rankers).
- **Relevant Coursework:** Quantum Computation & Information, Linear Algebra, Multivariable Calculus, Discrete Mathematics, Probability & Random Processes, Design & Analysis of Algorithms, Theory of Computation, Machine Learning.

## Research & Leadership Experience

**PESU Research Foundation (PESU-RF)** Bengaluru, India

*Research Intern – Quantum & Applied Machine Learning* *Oct. 2026 – Present*

- Investigating clinical aging dynamics on Longitudinal Ageing Study in India (LASI) data to formulate the multi-domain **Composite Aging Index (CAI)** encompassing 60+ clinical tests across physical, cognitive, biomarker, and mental health domains.
- Implemented generative multivariate Gaussian Copula models (Sklar's Theorem) for multi-domain data augmentation in the **IEEE AgeTech Global Dataset Competition**, preserving intricate inter-feature covariance structures within strict clinical limits.
- Designing parameterized quantum circuits to extract entanglement-based feature importance weights, benchmarking quantum-derived weights against classical PCA and autoencoder scoring models.

**$|Q\rangle$Forest (Quantum Computing Club, PES University)** Bengaluru, India

*Technical Head* *Aug. 2025 – Present*

- Leading the technical direction and research reading groups of PES University's official quantum computing student organization, mentoring 20+ undergraduate peers in quantum algorithms, quantum error correction, and quantum information theory.
- Organized hands-on technical tracks and hackathons utilizing **Qiskit** and **PennyLane**, covering Bell-CHSH inequalities, Grover's search, Quantum Phase Estimation (QPE), and Variational Quantum Eigensolvers (VQE).

**IEEE Standards Association (IEEE SA)** Bengaluru, India

*Research Intern – Experiential Program on Sustainability Standards (EPSS 2026)* *Jun. 2026 – Oct. 2026*

- Formulated an 8-rule verification framework specializing **ISO 14021** to validate corporate "Plastic Neutrality" claims across FMCG supply chains in India, addressing systemic vulnerabilities in Extended Producer Responsibility (EPR) compliance.
- Audited public sustainability disclosures and recycling declarations across 9 major FMCG conglomerates (HUL, Nestlé, ITC, P&G, etc.), identifying critical gaps including unverified cement kiln co-processing and double-counted EPR credits.
- Proposed a standardized JSON/CSV schema and serial-number-based retirement registry to eliminate double-counting ; conducted hybrid visits at Without® (Ashaya Recyclers), Toyoda Gosei, and CIPET.

## Selected Projects

**Qure: Quantum Kernel Genomic Classifier** | *Qiskit, PennyLane, Python, Scikit-learn* SIH Research Track

- Constructed a 16-qubit Quantum Support Vector Classifier (QSVC) utilizing a 2nd-order entangling **ZZFeatureMap** with phase gates and entangling CNOT $\rightarrow P(2(\pi - x_i)(\pi - x_j)) \rightarrow$ CNOT blocks to encode epistatic gene interactions into Hilbert space.
- Reduced 7,129 microarray gene expressions to a 16-gene biomarker panel using cross-validated mutual information selection; evaluated Fidelity Quantum Kernel estimation on IBM Quantum Aer statevector simulators under parameterized noise models.

**Quantum Kolmogorov-Arnold Networks (QKAN) & SineKAN** | *PyTorch, QML* Theoretical Study

- Reviewed foundational literature on Kolmogorov-Arnold representation theory and Quantum Kolmogorov-Arnold Networks (QKAN), studying the theoretical substitution of classical B-spline activation edges with parameterized quantum circuits (VQCs).
- Explored 1D SineKAN architectures in PyTorch utilizing Fourier basis functions as a classical precursor, analyzing the mathematical mapping from periodic sine activations to quantum single-qubit phase rotation gates.

## Technical Skills & Honors

**Quantum Computing**: Qiskit, PennyLane, Cirq, IBM Quantum Experience, QAOA, VQC, Quantum Kernel Methods

**Machine Learning & Math**: PyTorch, TensorFlow, Scikit-learn, Autoencoders, PCA, Gaussian Copula, SHAP, SMT/Z3, Linear Algebra, Convex Optimization, Sklar's Theorem, Statistical Hypothesis Testing

**Programming & Tools**: Python, C, C++, Linux/UNIX, Bash, Git, LaTeX, Docker, Jupyter

**Honors & Fellowships**: **IBM Qiskit Advocate** (Student Contributor & QGSS Mentor); **MRD Merit Scholarship** (Top Academic Ranker, PESU); **IEEE SA EPSS Fellow** (Sustainability & Technology Track 2026)